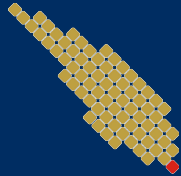




BMKG



ITERA

USER GUIDEBOOK & SOFTWARE TECHNICAL REFERENCE MANUAL

Undergraduate Internship Project of Geophysical Engineering Department

Faculty of Industrial Technology, Institut Teknologi Sumatera · Sleman Geophysical Station BMKG

OFFICIAL USER GUIDEBOOK & ENGINEERING TECHNICAL REFERENCE

BMKG Strong Motion Analyzer

(BSMA v2.0.0)

Integrated Computational Platform for Accelerograph Signal Processing, Seismic Kinematics, & Response Spectra

Operational Release Edition & Engineering Seismology Academic Reference

AUTHOR METADATA & ACADEMIC INTERNSHIP CREDENTIALS

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Live Cloud Web Application	: https://strong-motion.streamlit.app/
GitHub Source Repository	: https://github.com/ahmaddidan/BSMA-v.2

EXECUTIVE PROJECT SUMMARY & SYSTEM SPECIFICATIONS

1. PROJECT OVERVIEW & SOFTWARE CAPABILITIES

BMKG Strong Motion Analyzer (BSMA v2.0.0) is an integrated accelerograph signal analysis software suite independently designed and developed as part of an undergraduate Academic Internship by a student of the Department of Geophysical Engineering, Faculty of Industrial Technology, Institut Teknologi Sumatera (ITERA) at the Sleman Geophysical Station Class I, Meteorology, Climatology, and Geophysical Agency (BMKG) D.I. Yogyakarta (conducted July 20 – August 20, 2026). This software is an independent academic contribution and does not constitute an official operational system or commercial software product of BMKG.

The application automates the end-to-end processing workflow of strong ground motion records—spanning raw count ingestion, StationXML instrument transfer function deconvolution, multi-tier signal quality control (QC), digital signal processing (DSP), numerical kinematic integration (PGA, PGV, PGD), instrumental Modified Mercalli Intensity (MMI) estimation based on Worden et al. (2012) GMICE formulations, and 5% damped elastic Single-Degree-of-Freedom (SDOF) Pseudo-Spectral Acceleration (PSA) response spectra modeling featuring both the Nigam–Jennings (1969) analytical solver and the Newmark-Beta (1959) implicit integration solver, benchmarked against the Indonesian building code standard SNI 1726:2019.

Key Innovations: Fully documented open-source architecture, single-station and multi-station batch processing, interactive numerical SDOF solver cross-validation benchmarking, publication-grade multi-format reporting (PDF engineering reports, kinematic CSVs, spectral response matrices, and ZIP archives), and comprehensive execution provenance tracking ensuring scientific reproducibility.

2. SYSTEM SPECIFICATIONS & RUNTIME DEPENDENCIES

Operating System	Windows 10/11, Linux (Ubuntu 20.04+), macOS 12+
Python Runtime	Python 3.10 to 3.13 (Primary benchmark environment: Python 3.13.2 64-bit)
GUI & Visualization Stack	Streamlit (≥1.35.0), Plotly (≥5.20.0), Matplotlib (≥3.8.0)
Scientific Computing & DSP	ObsPy (≥1.4.0), NumPy (≥1.26.0), SciPy (≥1.12.0), Pandas (≥2.0.0)
Document Generation	PyMuPDF / Fitz (≥1.24.0), FPDF (≥1.7.2)
Automated Test Suite	Pytest (≥8.0.0) — 83 unit and regression benchmark tests (100% passing)
Memory Requirements	Minimum 4 GB RAM (8 GB recommended for multi-station batch workflows)
Deployment Options	Streamlit Community Cloud & Local Workstation Desktop Server

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CHAPTER I: INTRODUCTION & TECHNICAL SCOPE

1.1 Background & Critical Need for Strong-Motion Analysis

Indonesia is located within one of the most seismotectonically complex regions in the world, formed by the convergence of four major tectonic plates (Eurasia, Indo-Australia, Pacific, and Philippine Sea). Accelerograph records of strong ground motion provide indispensable, unsaturated records of ground acceleration during moderate-to-large earthquakes, serving as the cornerstone for seismic hazard mitigation, rapid shake mapping (USGS ShakeMap), and structural earthquake engineering.

1.2 Development Objectives & Design Philosophy

The **BSMA v2.0.0** platform was designed to address key engineering and research challenges:

- **End-to-End Automation:** Seamlessly ingests raw records, performs instrument response correction, evaluates signal quality, computes kinematics, and calculates response spectra within a unified interface.
- **Scientific Provenance & Reproducibility:** Every algorithm and parameter conforms to established literature (Boore & Bommer 2005, Worden et al. 2012, Nigam & Jennings 1969) and is logged into an execution audit trail.
- **Dual Accessibility:** Accessible both as a zero-installation cloud web application ([strong-motion.streamlit.app](#)) and as a local high-performance workstation suite.

1.3 Functional Scope & Module Matrix

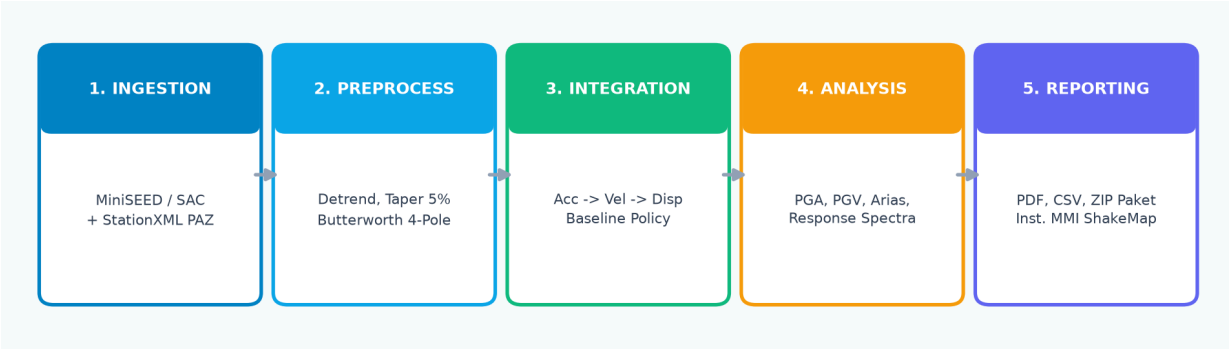
The BSMA v2.0.0 suite is organized into 7 primary functional analysis modules:

Module / Tab	Primary Scope	Key Scientific Deliverables
1. Summary	Executive station overview & waveform preview	Station coordinates, triaxial peaks, QC status badge, MMI badge
2. Waveforms	Interactive 3-component kinematics (a, v, d)	Synchronized plots for Acceleration (m/s²), Velocity (cm/s), Displacement (cm)
3. Quality Control	Signal integrity diagnostics & anomaly screening	Quality Score (0–100), polar metric radar, SNR (dB), PASS/WARN/FAIL status
4. Strong Motion	Seismic energy accumulation & duration	Husid plot of Arias Intensity (Ia), Significant Duration intervals D5-95 / D5-75
5. Intensity	Instrumental ground shaking classification	USGS ShakeMap MMI (Worden et al., 2012), perceived shaking & damage cards
6. Spectrum	Elastic SDOF 5% damped response spectra	PSA curves (T = 0.01–10 s), SNI 1726:2019 design overlay, SDOF Benchmark
7. Report	Publication-grade reporting & multi-format export	Comprehensive technical PDF report, summary CSV, spectral matrix CSV, ZIP

[REFERENCE STANDARD] Academic Integrity Standard

This software was independently developed as an undergraduate innovation project during an academic internship at Sleman Geophysical Station BMKG. All mathematical formulations and visualization routines have been thoroughly verified against an 83-test automated test suite using real operational BMKG accelerograms.

CHAPTER II: PIPELINE ARCHITECTURE & DATA INGESTION



2.1 Sequential Signal Processing Architecture

The BSMA architecture enforces a rigorous 5-stage sequential execution pipeline designed for full data provenance tracking and scientific reproducibility. Every processing step records filter parameters, algorithm choices, and execution timestamps into an audit log.

2.2 Supported Waveform Data Formats (MiniSEED & SAC)

BSMA natively ingests the two dominant formats in international earthquake seismology:

- **MiniSEED (.mseed)**: The FDSN standard format for compressed digital seismic time series, supporting multi-channel triaxial streams with microsecond UTC timing precision.
- **SAC (.sac)**: The IRIS Seismic Analysis Code format, supporting detailed header metadata (station coordinates, channel orientation, and sampling rate).

2.3 StationXML Instrument Response Deconvolution

When input waveforms are recorded in raw digital counts, BSMA applies instrument transfer function deconvolution via **StationXML (.xml)**:

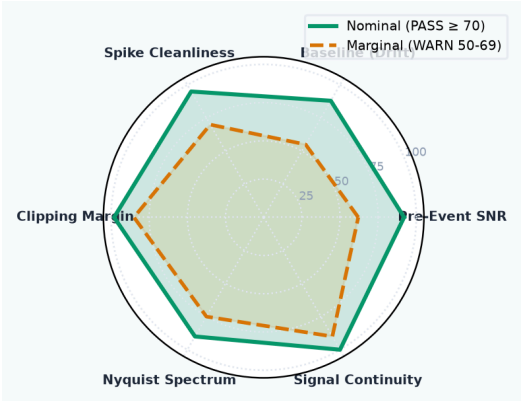
- Parses poles, zeros (PAZ), and stage sensitivity gains via ObsPy.
- Converts digital counts into true physical ground acceleration in m/s^2 or Gal (cm/s^2).
- **Physical Acceleration Bypass Mode**: If uploaded records are already calibrated in physical acceleration units, the system automatically detects this and bypasses deconvolution, preventing double-correction distortion.

CHAPTER III: QUALITY CONTROL & SIGNAL INTEGRITY

3.1 Three-Tier Quality Evaluation

BSMA provides an automated signal screening framework producing an objective Quality Score (0–100):

- **QC PASS (Score ≥ 70):** High-fidelity record suitable for structural engineering and spectra.
- **QC WARNING (Score 50–69):** Non-fatal anomalies (marginal SNR 10–20 dB, isolated spikes, mild baseline tilt). Filtering adjustments advised.
- **QC FAIL (Score < 50):** Fatal anomalies detected (sensor clipping, ADC saturation, flatlines, or corrupted channels).



3.2 Fatal Overrides: Sensor Clipping & ADC Saturation

When amplitude truncation (sensor clipping exceeding full-scale range) or bit saturation (repeated identical maximum integer counts) is detected, the system triggers a mandatory **Fatal Override**, forcing the status to **FAIL** regardless of other metrics. Clipped records are strictly invalidated for peak ground motion analysis because PGA is truncated and MMI estimation becomes severely underestimated.

3.3 Standard Six-Class Diagnostic Taxonomy

Class	Physical Characteristics	QC Status	Operational Implication
Class 1	Nominal Strong Motion	PASS (≥ 70)	Clean record; PGA, PGV, PGD, MMI, and response spectra fully reliable.
Class 2	Low SNR / Weak Motion	WARNING (50-69)	Weak shaking; highpass filter corner adaptively raised (0.20–0.40 Hz).
Class 3	Baseline Offset / Drift	WARNING (50-69)	DC shift detected; polynomial detrending applied prior to integration.
Class 4	Glitches & Spikes	WARNING / FAIL	Transient electrical/mechanical spike; manual verification advised.
Class 5	Sensor Clipping / Saturation	FAIL (< 50 Fatal)	Peaks clipped; strictly invalid for engineering design spectra.
Class 6	Sensor Flatline / Dead Channel	FAIL (< 50 Fatal)	Dead sensor or broken transmission link; record disqualified.

CHAPTER IV: DIGITAL SIGNAL PROCESSING & FILTERING

4.1 Baseline Detrending (Mean & Polynomial Drift Removal)

Prior to filtering, acceleration time series are cleansed of zero-frequency offsets (DC bias) and mechanical drift following the benchmark guidelines of **Boore & Bommer (2005)**. Without initial detrending, uncorrected baseline shifts propagate quadratically through double integration, producing catastrophic parabolic displacement trajectories.

4.2 5% Tukey Cosine Window (Tapering & Spectral Leakage Mitigation)

Applying digital filters to finite-length time series induces edge discontinuities. BSMA applies a **5% Tukey window** to both record boundaries in accordance with **Harris (1978)**:

- Mitigates high-frequency spectral leakage across the discrete Fourier transform.
- Ensures smooth asymptotic transitions to zero at the record edges without altering the primary strong-motion phase amplitudes.

4.3 Zero-Phase 4th-Order Butterworth Bandpass Filtering (Forward-Backward sosfiltfilt)

Filtering is performed using a **4th-order Butterworth bandpass filter** implemented via two-pass forward-backward processing (`scipy.signal.sosfiltfilt`):

- **Zero Net Phase Lag ($\Delta \phi = 0^\circ$)**: Eliminates phase distortion entirely, guaranteeing that phase arrival times and peak accelerations remain strictly locked to their exact physical timestamps.
- **Effective 48 dB/octave Attenuation Rate**: Forward-backward filtering squares the magnitude transfer function ($|H(f)|^2$), achieving an 8th-order stopband rolloff while preserving the intrinsic numerical stability of the 4th-order prototype (Virtanen et al., 2020).
- **Second-Order Sections (SOS)**: Prevents numerical precision loss and pole sensitivity issues inherent in high-order polynomial representations.

4.4 Nyquist Safeguard Limit (0.40 fs) & Adaptive Low-Frequency SNR Floor

- **Nyquist Safeguard**: The upper cutoff frequency is strictly capped at $f_{\max} \leq 0.80 f_{\text{Nyquist}} = 0.40 f_s$ to safeguard against digital aliasing artifacts.
- **Adaptive SNR Floor**: For records with marginal pre-event signal-to-noise ratios ($\text{SNR} < 20 \text{ dB}$ or $< 10 \text{ dB}$), the highpass corner $f_{\min}$ is automatically elevated from 0.10 Hz to 0.20 Hz or 0.40 Hz, shielding subsequent kinematic integrations from low-frequency drift.

CHAPTER V: KINEMATIC INTEGRATION & BASELINE POLICY

5.1 Cumulative Trapezoidal Numerical Integration ($a(t) \rightarrow v(t) \rightarrow d(t)$)

Velocity time histories $v(t)$ and displacement time histories $d(t)$ are evaluated from the corrected acceleration $a(t)$ via sequential cumulative trapezoidal numerical integration:

$$v(t) = \int_0^t a(\tau) \, d\tau \approx \sum_{i=1}^k \frac{a(t_i) + a(t_{i-1})}{2} \Delta t$$

$$d(t) = \int_0^t v(\tau) \, d\tau \approx \sum_{i=1}^k \frac{v(t_i) + v(t_{i-1})}{2} \Delta t$$

5.2 Strict Scientific Baseline Policy (Acceleration-Only Correction)

In strict adherence to the principles of kinematic calculus and modern earthquake engineering standards, BSMA enforces a **Strict Baseline Correction Policy**:

- **Exclusively Applied to Acceleration:** Detrending, tapering, and bandpass filtering are completed entirely on acceleration *prior* to integration.
- **No Post-Integration Baseline Forcing:** BSMA deliberately avoids applying artificial polynomial or highpass filtering post-integration to $v(t)$ or $d(t)$, strictly preserving true derivative consistency ($a = \dot{v} = \ddot{d}$). Post-integration baseline forcing introduces non-physical phase distortions and artificially diminishes actual seismic peak amplitudes.

5.3 Physical Interpretation of PGD (Transient Dynamic Peak vs. Static Fling-Step)

The Peak Ground Displacement (PGD) output by BSMA represents the **transient dynamic peak displacement** within the passband of the digital filter ($f_{\min} - f_{\max}$), rather than the permanent tectonic static dislocation (*static fling-step*):

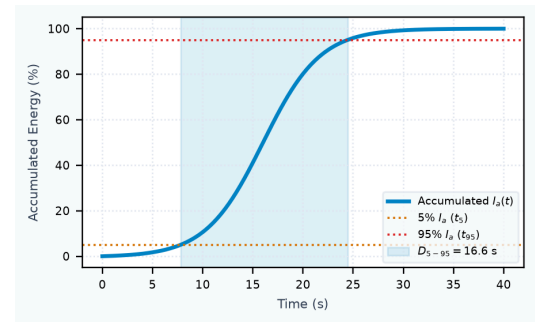
- Resolving permanent near-fault tectonic displacement seismologically requires specialized nonlinear baseline correction (e.g., Iwan et al. 1985) or co-located high-rate GNSS instrumentation.
- Transient dynamic PGD is the engineering metric governing peak structural drift, inter-story shear strain, and building damage.

CHAPTER VI: KINEMATIC PARAMETERS & SEISMIC ENERGY

6.1 Peak Ground Motion Extraction

BSMA extracts absolute peak ground motion parameters:

- **PGA:** $\max_t |a(t)|$ in m/s^2 or Gal (cm/s^2).
- **PGV:** $\max_t |v(t)|$ in cm/s .
- **PGD:** $\max_t |d(t)|$ in cm .
- **$V_{\max}/A_{\max}$ Ratio:** Indicator of predominant spectral period and source-to-site distance.



6.2 Cumulative Arias Intensity (I_a) & Husid Plot

Arias Intensity (I_a) quantifies the total seismic shaking energy dissipated into a population of structural oscillators over the entire duration of ground motion, formulated by **Arias (1970)**:

$$I_a = \frac{\pi}{2g} \int_0^{t_{\max}} [a(t)]^2 dt \quad [\text{m/s}]$$

The **Husid Plot** displays the normalized cumulative buildup of Arias Intensity over time (0% to 100%), visualizing the instantaneous energy arrival rate and the most destructive shaking pulses.

6.3 Significant Duration Intervals (D_{5-95} & D_{5-75})

Strong motion duration is evaluated using the energy accumulation boundaries defined by **Trifunac & Brady (1975)**:

- **Significant Duration D_{5-95} :** $t_{95} - t_5$ (time interval between 5% and 95% cumulative Arias energy). This metric governs cyclic structural fatigue degradation, pore-water pressure generation, and soil liquefaction triggering.
- **Significant Duration D_{5-75} :** $t_{75} - t_5$ (time interval between 5% and 75% cumulative Arias energy), widely applied in nonlinear hysteretic dissipation and ductility demand assessments.

CHAPTER VII: INSTRUMENTAL MMI INTENSITY ESTIMATION

7.1 Worden et al. (2012) GMICE Formulations (USGS ShakeMap)

Instrumental Modified Mercalli Intensity (MMI) is computed using the global standard Ground-Motion Intensity Conversion Equations (GMICE) established by **Worden et al. (2012)**, which forms the core of the USGS ShakeMap operational framework.

Formulation Based on PGA (Gal):

$$MMI_{PGA} = \begin{cases} 1.78 + 1.55 \log_{10}(PGA), & \text{if } 1.57 \leq PGA \leq 1.60 \\ -1.60 + 3.70 \log_{10}(PGA), & \text{if } PGA > 1.57 \end{cases}$$

Formulation Based on PGV (cm/s):

$$MMI_{PGV} = \begin{cases} 3.78 + 2.99 \log_{10}(PGV), & \text{if } 0.53 \leq PGV \leq 2.40 \\ 2.40 + 4.96 \log_{10}(PGV), & \text{if } PGV > 0.53 \end{cases}$$

7.2 Maximum Horizontal Component (Max-H) & PGV Dominance Transition

- **Reference Component:** Peak ground motions are evaluated strictly on the **Maximum Horizontal Component (Max-H)** (envelope of North-South and East-West), excluding the vertical channel Z/U in accordance with earthquake engineering convention.
- **PGV Dominance at Strong Shaking:** At moderate-to-severe ground shaking levels ($MMI \geq 5.0$), the PGV relationship automatically governs final intensity because ground velocity correlates directly with kinetic energy flux and building structural damage.

7.3 MMI Scale Correlation, Perceived Shaking, & Structural Damage Matrix

MMI	PGA (Gal)	PGV (cm/s)	Perceived Shaking	Potential Physical Structural Damage
I	< 0.17	< 0.1	Not Felt	No damage whatsoever.
II - III	0.17 - 1.4	0.1 - 1.1	Weak / Light	Felt by resting persons indoors; no structural damage.
IV	1.4 - 3.9	1.1 - 3.4	Moderate	Dishes and windows rattle; walls crack slightly; no structural failure.
V	3.9 - 9.2	3.4 - 8.1	Strong	Felt by nearly everyone; small unstable objects overturned.
VI	9.2 - 18	8.1 - 16	Very Strong	Minor non-structural damage; fallen plaster and masonry cracks.
VII	18 - 34	16 - 31	Severe	Moderate damage in standard masonry structures; slight in well-designed buildings.
VIII	34 - 65	31 - 60	Violent	Heavy structural damage; partial collapse in unreinforced masonry.
IX+	> 65	> 60	Extreme	Catastrophic structural failure; widespread building collapses and foundation shifts.

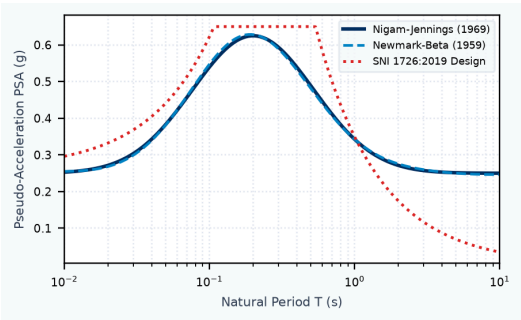
CHAPTER VIII: SDOF RESPONSE SPECTRA & SNI 1726:2019 CODE

8.1 Elastic SDOF 5% Damped Oscillator Modeling

The Pseudo-Spectral Acceleration (PSA) spectrum models the peak dynamic displacement response $u(t)$ of an array of Single-Degree-of-Freedom (SDOF) linear elastic oscillators with 5% critical damping ($\xi = 0.05$):

$$\text{PSA}(T, \xi) = \omega_d^2 S_d(T, \xi) = \omega_d^2 \max_t |u(t)| \quad [g \text{ or } \text{m/s}^2]$$

Evaluated across 150 logarithmically spaced natural periods $T = 0.01 - 10.0$ s.



8.2 Dual Numerical Solver Engines: Nigam-Jennings vs. Newmark-Beta

BSMA equips researchers with two independent numerical engines:

- **Nigam–Jennings (1969):** Exact analytical state-transition matrix formulation assuming piecewise-linear ground excitation. Highly superior in computational precision for short periods ($T < 0.1$ s).
- **Newmark-Beta (1959):** Step-by-step implicit time integration using the average acceleration scheme ($\gamma = 1/2, \beta = 1/4$), providing unconditional numerical stability across all periods.

8.3 SDOF Solver Cross-Validation & Numerical Benchmarking

An interactive verification tool computes maximum relative deviation, mean relative deviation, and RMS error between solvers in real time. Mean solver deviation remains strictly $< 5\%$, rigorously confirming the high-frequency rigid limit anchor $\lim_{T \rightarrow 0} \text{PSA}(T) = \text{PGA}$.

8.4 SNI 1726:2019 Building Code Design Spectrum Overlay

BSMA enables direct overlay comparison against the Indonesian building code standard **SNI 1726:2019** design spectra ($S_{DS}, S_{D1}, T_0, T_{s0}$) across soil site classes (Hard Rock SA/SB, Medium Soil SD, and Soft Soil SE). This functions as an engineering reference overlay to verify whether observed shaking demands exceed standard building design capacities.

CHAPTER IX: GUI OPERATIONAL GUIDE & ANALYSIS WORKFLOW

9.1 Single-Station Analysis Workflow

1. **Data Ingestion (Sidebar):** Upload triaxial MiniSEED (.mseed) or SAC (.sac) waveform files. Optionally attach StationXML (.xml) metadata for instrument deconvolution.
2. **Processing Configuration:** Define bandpass filter corners ($f_{\min} = 0.10$ Hz, $f_{\max} = 25.0$ Hz), detrending polynomial order, and SDOF response spectrum solver engine.
3. **Interactive Tab Exploration:**
 - *Tab 1 (Summary):* Review station geographic coordinates, 3-component waveform preview, and QC health badges.
 - *Tab 2 (Waveforms):* Interactively explore synchronized time histories for acceleration, velocity, and displacement.
 - *Tab 3 (Quality Control):* Inspect noise metric radar charts, Power Spectral Density (PSD), and SNR diagnostics.
 - *Tab 4 (Strong Motion):* Monitor Arias Intensity Husid accumulation curves, D_{5-95} duration, and $V_{\max}/A_{\max}$.
 - *Tab 5 (Intensity):* Examine instrumental MMI classifications (Worden et al., 2012) and perceived damage cards.
 - *Tab 6 (Spectrum):* Visualize elastic 5% damped PSA response curves, SNI 1726:2019 overlays, and solver benchmarks.

9.2 Multi-Station Batch Processing Workflow

For regional earthquake events recorded across extensive accelerograph networks: • Upload multiple station files simultaneously via the batch ingestion panel.

- The system executes an asynchronous, isolated pipeline loop with automatic failure isolation.
- Generates a unified **Regional Comparison Matrix** sortable by highest PGA or maximum instrumental MMI.

9.3 Reporting & Multi-Format Data Export (Tab 7 Report)

BSMA provides comprehensive data products for post-earthquake reporting and research:

- **Technical PDF Report:** Publication-grade engineering report featuring full seismograms, kinematic tables, and audit logs.
- **Kinematics Summary CSV:** Tabular spreadsheet containing peak kinematics (PGA, PGV, PGD, MMI, D_{5-95} , I_a).
- **Spectral Response Matrix CSV:** Discrete period (T) versus PSA values for all three orthogonal components.
- **Full Export ZIP Archive:** Bundled archive containing the PDF report, all CSV tables, and processed waveform series.

CHAPTER X: TROUBLESHOOTING GUIDE & OPERATIONAL LIMITATIONS

10.1 Frequently Asked Questions (Troubleshooting FAQ)

• **Q: Why does the displacement trajectory curve drastically upward or downward?**

A: Caused by residual baseline drift during double numerical integration of low-frequency noise. Solution: Increase the highpass cutoff frequency ($f_{\min}$) from 0.05 Hz to 0.10 Hz or 0.20 Hz in the filter panel.

• **Q: Why do I receive the notification 'StationXML Response Correction Bypassed'?**

A: The StationXML file does not match the station/network code of the record, or input waveforms are already calibrated in physical acceleration (m/s^2). The system safely defaults to Physical Acceleration mode. Solution: Ensure station/channel metadata match exactly.

• **Q: Why does the accelerograph PGA differ from a broadband seismometer at the same site?**

A: Accelerographs are designed for high-amplitude strong ground motions without clipping, whereas broadband sensors record weak-motion velocities. For structural engineering and MMI estimation, accelerographs are the standard reference.

• **Q: Why does Quality Control report WARNING or FAIL even when the waveform looks clear?**

A: The QC engine evaluates 6 distinct diagnostic metrics (clipping, spikes, flatlines, pre-event noise, SNR, DC offset). WARNING is typically triggered by a pre-event window < 5 seconds or SNR < 10 dB. Inspect Tab 3 for detailed penalty breakdowns.

• **Q: When should I select Nigam-Jennings instead of Newmark-Beta?**

A: Nigam-Jennings (1969) is strongly recommended for short-period spectra ($T < 0.1$ s) due to its exact analytical piecewise-linear formulation. Newmark-Beta (1959) uses implicit average acceleration integration. Both exhibit mean relative discrepancy < 5% verified in Tab 6.

10.2 Methodological & Operational Limitations

- **PGD Sensitivity:** Double numerical integration is inherently sensitive to residual low-frequency baseline drift.
- **StationXML Dependency:** Deconvolution accuracy is strictly bounded by the validity of sensor PAZ and sensitivity gains.
- **High Noise Records:** Records with estimated SNR < 3 dB are not recommended for quantitative kinematic modeling.
- **Instrumental MMI:** Represents an empirical kinematic approximation and does not replace field macroseismic surveys.

10.3 Cloud Web Access & GitHub Source Code

- **Live Cloud Web Application:** <https://strong-motion.streamlit.app/> (Zero-installation cloud access).
- **GitHub Source Repository:** <https://github.com/ahmaddidan/BSMA-v.2> (Full open-source suite & automated tests).

REFERENCES & SCIENTIFIC BIBLIOGRAPHY

[TECHNICAL NOTE] Theoretical Foundations & Computational Framework

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DEVELOPER PROFILE & ACADEMIC ENDORSEMENT

1. SOFTWARE DEVELOPER PROFILE

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2. ACADEMIC INTEGRITY STATEMENT & DOCUMENT ENDORSEMENT

Statement of Scientific Integrity:

This User Guidebook and Technical Reference Manual is compiled as the official documentation for the **BMKG Strong Motion Analyzer (BSMA v2.0.0)** software suite in fulfillment of the undergraduate Academic Internship requirements of the Geophysical Engineering Study Program ITERA at Sleman Geophysical Station Class I, BMKG D.I. Yogyakarta. All mathematical formulations, numerical integration routines, structural dynamics solvers, and graphical interfaces have been rigorously validated against actual BMKG accelerograms and verified against closed-form analytical synthetic solutions.

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